Convolutional neural networks. A convolutional layer slides a small learned filter across an image and computes a dot product at each position, producing a feature map. Because the same filter weights are reused at every position, the layer has far fewer parameters than a fully connected layer and is translation equivariant. Stacking convolutional layers lets the network build features from edges to textures to object parts. Pooling layers reduce the spatial size of the feature maps. Max pooling keeps the largest value in each window. CNNs were the dominant approach to image classification after the 2012 ImageNet competition, when AlexNet, a deep convolutional network trained on GPUs, reduced the top-5 error rate substantially compared with the previous best entries.

Recurrent networks and attention. Recurrent neural networks process a sequence one element at a time and carry a hidden state from step to step. Plain recurrent networks struggle with long sequences because gradients shrink or explode over many steps. The long short-term memory network, or LSTM, adds a cell state and three gates, the forget, input, and output gates, which control what information is kept, written, and read. The gated recurrent unit, or GRU, is a simpler design with two gates. Transformers replace recurrence with self-attention, in which every position computes a weighted sum over all other positions. The weights come from scaled dot products between queries and keys. Because attention does not process tokens in order, transformers use positional encodings to represent the order of the sequence, and they can be trained in parallel across the whole sequence.